INITIAL STATIC VERTICAL PILE LOAD TEST

ON 1000MM DIA PILE

FOR Prestige Nautilus Worli
AT Rehab SRA Plot B

(TEST PILE TP-01)

Site Image

Client:	Prestige
Contractor:	L&T
Test Date:	06 January 2026
Report No:	IVPLT-03

Test conducted as per IS 2911 (Part 4) - 2013

CONTENTS

1.0 General

1.1 Prestige decided to carry out static pile testing work on 1000mm diameter pile to estimate load carrying capacity in vertical direction and settlement. This is the Initial Vertical Pile Load Test at Rehab SRA Plot B.

1.2 This report covers data for one vertical pile load test. This report covers calculation of safe load capacity for pile based on data collected during fieldwork.

1.3 The following codes of practices have been adopted:

- IS 2911 (Part 4) - 2013 "Code of Practice for Design and Construction of Pile Foundations - Load Tests on Piles"
- IS 14593 - 1998 (Reaffirmed 2003) "Design and Construction of Bored Cast-in-Situ Piles Founded on Rocks – Guidelines"

2.0 Scope of Work

Pile details are tabulated as below.

2.1 Pile Details for Initial Pile (Vertical Load Test)

Location	**Rehab SRA Plot B**
Pile ID	**TP-01**
Pile Diameter	**1000 mm**
Pile Depth	**10 m**
Concrete Grade	**M25**
Maximum Vertical Safe Capacity	**1250 MT**

2.2 Vertical Test Load for Initial Pile

The design vertical load on the pile is **1250 MT**.

The pile is required to be tested to a load of **3125 MT** (2.5 × design load).

2.3 Equipment Details

Hydraulic Jack	**As per site specifications**
Ram Area	**5102 cm^2**
Dial Gauge Least Count	**0.01 mm**

3.0 Methodology

The Initial Vertical Pile Load Test was conducted in accordance with IS 2911 (Part 4) - 2013 "Code of Practice for Design and Construction of Pile Foundations - Load Test on Piles".

3.1 Test Setup

A hydraulic jack of adequate capacity was used to apply the load, reacting against a sturdy reaction frame. Four dial gauges (0.01 mm least count) were fixed on the pile head to record settlements.

3.2 Test Procedure

1. Load was applied in increments of approximately 20% of design load (250.0 MT) up to the test load of 3125 MT.
2. Each load increment was maintained until the rate of settlement was less than 0.1mm/hour, or for a minimum of 1 hour.
3. Maximum load was maintained for 24 hours as per IS 2911 requirements for initial tests.
4. Load was released in the same increments as the loading phase.
5. Settlement readings were recorded at each load increment using four dial gauges positioned at 90° intervals.

3.3 Acceptance Criteria (IS 2911 Part 4)

As per IS 2911 (Part 4) - 2013, the acceptance criteria for Initial Vertical Pile Load Test are:

1. Net settlement at design load shall not exceed 12mm or 2% of pile diameter (20.0mm), whichever is less.
2. Safe load shall be taken as two-thirds of the load at which total settlement equals 12mm.
3. Safe load shall not exceed half the load at which total settlement equals 10% of pile diameter (100.0mm).

3.4 Load Increment Summary

Jack Ram Area: **5102 cm²**

Design Load: **1250 MT**

Test Load (2.5× Design): **3125 MT**

Load Increment (20% of Design): **250.0 MT**

Note: As the least count of the pressure gauge is limited, exact incremental loads may not be attained. Hence, values close to the incremental load are considered.

Table 1 – Load Sequence

The sequence of loading and unloading shall be as described below for **IVPLT on 1000mm Dia Pile (1250 MT Design Load)**.

SR. NO.	PRESSURE (KG/CM²)	LOAD (MT)	READING TIME
LOADING PHASE			
1	0	0.00	0
2	49	250.00	1, 15, 30, 45, 60 mins
3	98	500.00	1, 15, 30, 45, 60 mins
4	147	750.00	1, 15, 30, 45, 60 mins
5	196	1000.00	1, 15, 30, 45, 60 mins
6	245	1250.00	1, 15, 30, 45, 60 mins
7	294	1500.00	1, 15, 30, 45, 60 mins
8	343	1750.00	1, 15, 30, 45, 60 mins
9	392	2000.00	1, 15, 30, 45, 60 mins
10	441	2250.00	1, 15, 30, 45, 60 mins
11	490	2500.00	1, 15, 30, 45, 60 mins
12	539	2750.00	1, 15, 30, 45, 60 mins
13	588	3000.00	1, 15, 30, 45, 60 mins
14	**613**	**3125.00**	**24 hours (1440 mins)**

SR. NO.	PRESSURE (KG/CM²)	LOAD (MT)	READING TIME
UNLOADING PHASE			
15	564	2875.00	1, 5, 15 mins
16	515	2625.00	1, 5, 15 mins
17	466	2375.00	1, 5, 15 mins
18	417	2125.00	1, 5, 15 mins
19	368	1875.00	1, 5, 15 mins
20	319	1625.00	1, 5, 15 mins
21	270	1375.00	1, 5, 15 mins
22	221	1125.00	1, 5, 15 mins
23	172	875.00	1, 5, 15 mins
24	123	625.00	1, 5, 15 mins
25	74	375.00	1, 5, 15 mins
26	25	125.00	1, 5, 15 mins
27	0	0.00	1, 5, 15 mins

4.0 Results

4.1 Test Results Summary

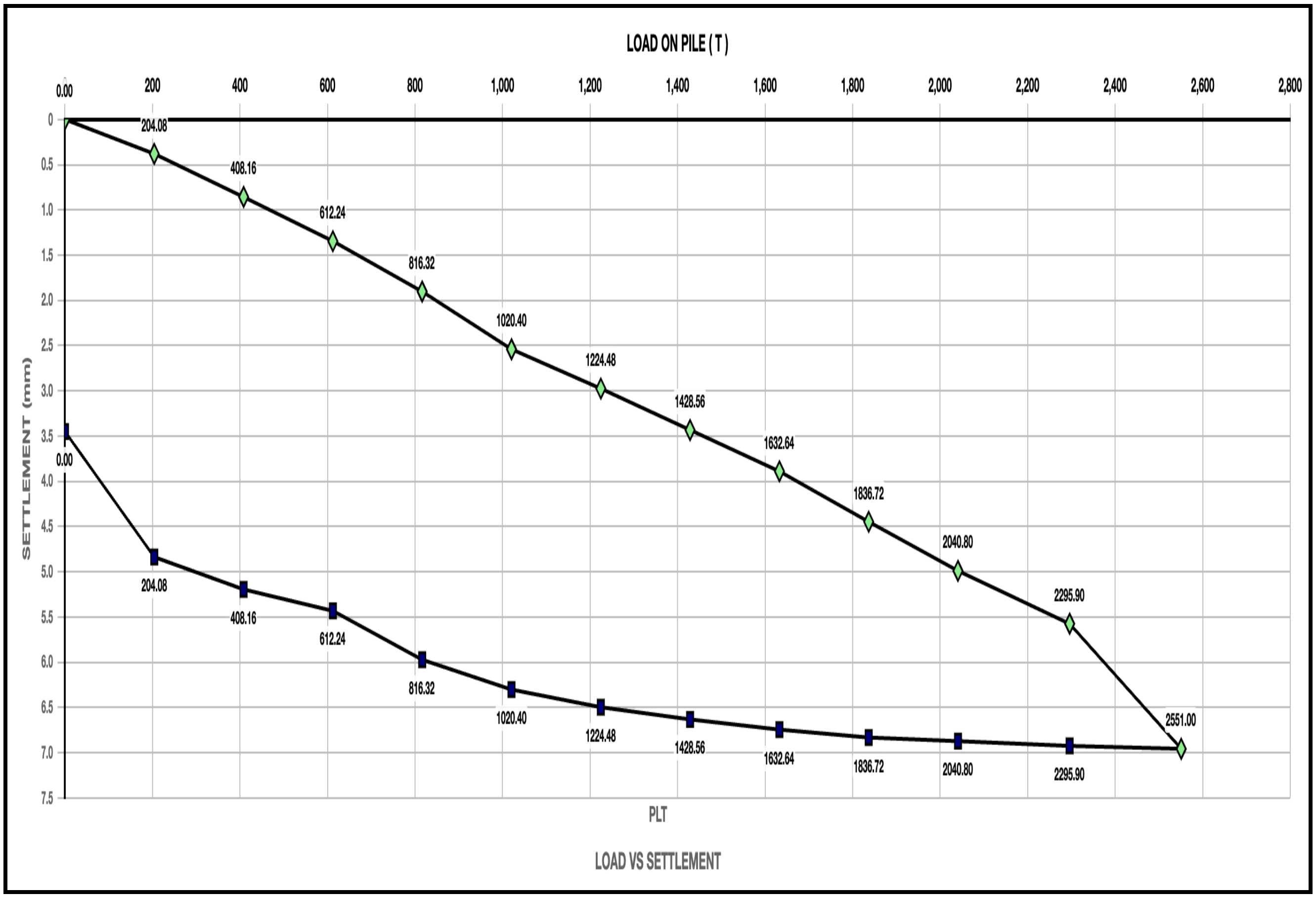
Maximum Settlement Recorded	**7.05 mm**
Elastic Rebound	**3.58 mm**
Net Settlement (Max - Rebound)	**3.47 mm**
Settlement Limit (IS 2911)	**12 mm**
Safe Load Adopted	**1250.0 MT**
Governing Criterion	**DESIGN**
Test Status	**PASSED**

4.2 Assessment

The net settlement of 3.47 mm is within the permissible limit of 12 mm as per IS 2911 (Part 4) - 2013.

The safe load adopted for the pile is **1250.0 MT** based on the design criterion.

5.0 Load vs Settlement Curve



Maximum Settlement at 2551.00 T: 7.05 mm

TEST LOAD	MAX SETTLEMENT	NET SETTLEMENT	TOTAL REBOUND
3125.0 MT	7.05 mm	3.47 mm	3.58 mm

TEST PASSED ✓

Net settlement 3.47mm is within the 12mm limit (IS 2911 Part 4)

5.1 Record of Pile Load Test

LOADING

Load (T)	Average Deflection (mm)
0.00	0.00
204.08	0.38
408.16	0.86
612.24	1.34
816.32	1.90
1020.40	2.54
1224.48	2.98
1428.56	3.43
1632.64	3.89
1836.72	4.45
2040.80	4.99
2295.90	5.58
2551.00	6.96

UNLOADING

Load (T)	Average Deflection (mm)
2295.90	6.92
2040.80	6.87
1836.72	6.83
1632.64	6.75
1428.56	6.63
1224.48	6.50
1020.40	6.30
816.32	5.97
612.24	5.43
408.16	5.20
204.08	4.84
0.00	3.45

6.0 Conclusion

Based on the Initial Vertical Pile Load Test conducted on pile TP-01 at Rehab SRA Plot B, the following observations and conclusions are drawn:

1. The pile was loaded up to 3125.0 MT (2.5 times the design load of 1250 MT) as per IS 2911 (Part 4) - 2013.
2. The maximum settlement recorded at test load was 7.05 mm.
3. After complete unloading, the elastic rebound was 3.58 mm.
4. The net settlement (residual) is 3.47 mm, which is within the permissible limit of 12 mm.
5. The safe load adopted for the pile is 1250.0 MT based on the design criterion.
6. **The pile HAS PASSED the Initial Vertical Pile Load Test as per IS 2911 (Part 4) - 2013.**

Therefore, the design safe load of 1250 MT can be adopted as the working load capacity for this pile.

7.0 Observation Sheet

Load increment readings as per IS 2911 (Part 4)

DATE	TIME (Hrs)	PRESSURE GAUGE READING kg/cm²	LOAD IN MT	Dial Gauge				AVERAGE SETTLEMENT IN MM	REMARK
				Reading 1	Reading 2	Reading 3	Reading 4		
Loading									
-	-	-	0.00	-	-	-	-	0.00	-
	-	-	204.08	-	-	-	-	0.37	-
	-	-		-	-	-	-	0.37	-
	-	-		-	-	-	-	0.38	-
	-	-		-	-	-	-	0.38	-
	-	-		-	-	-	-	0.38	-
	-	-	408.16	-	-	-	-	0.84	-
	-	-		-	-	-	-	0.84	-
	-	-		-	-	-	-	0.85	-
	-	-		-	-	-	-	0.86	-
	-	-		-	-	-	-	0.86	-
	-	-	612.24	-	-	-	-	1.32	-
	-	-		-	-	-	-	1.33	-
	-	-		-	-	-	-	1.33	-
	-	-		-	-	-	-	1.34	-
	-	-		-	-	-	-	1.34	-
	-	-	816.32	-	-	-	-	1.85	-
	-	-		-	-	-	-	1.86	-
	-	-		-	-	-	-	1.88	-
	-	-		-	-	-	-	1.89	-

DATE	TIME (Hrs)	PRESSURE GAUGE READING kg/cm²	LOAD IN MT	Dial Gauge				AVERAGE SETTLEMENT IN MM	REMARK
				Reading 1	Reading 2	Reading 3	Reading 4		
	-	-		-	-	-	-	1.90	-
	-	-	1020.40	-	-	-	-	2.50	-
	-	-		-	-	-	-	2.51	-
	-	-		-	-	-	-	2.52	-
	-	-		-	-	-	-	2.52	-
	-	-		-	-	-	-	2.54	-
	-	-	1224.48	-	-	-	-	2.93	-
	-	-		-	-	-	-	2.94	-
	-	-		-	-	-	-	2.95	-
	-	-		-	-	-	-	2.96	-
	-	-		-	-	-	-	2.98	-
	-	-	1428.56	-	-	-	-	3.39	-
	-	-		-	-	-	-	3.39	-
	-	-		-	-	-	-	3.41	-
	-	-		-	-	-	-	3.42	-
	-	-		-	-	-	-	3.43	-
	-	-	1632.64	-	-	-	-	3.85	-
	-	-		-	-	-	-	3.85	-
	-	-		-	-	-	-	3.86	-
	-	-		-	-	-	-	3.88	-
	-	-		-	-	-	-	3.89	-
	-	-	1836.72	-	-	-	-	4.40	-
	-	-		-	-	-	-	4.40	-
	-	-		-	-	-	-	4.41	-
	-	-		-	-	-	-	4.43	-
	-	-		-	-	-	-	4.45	-

DATE	TIME (Hrs)	PRESSURE GAUGE READING kg/cm²	LOAD IN MT	Dial Gauge				AVERAGE SETTLEMENT IN MM	REMARK
				Reading 1	Reading 2	Reading 3	Reading 4		
	-	-	2040.80	-	-	-	-	4.93	-
	-	-		-	-	-	-	4.95	-
	-	-		-	-	-	-	4.97	-
	-	-		-	-	-	-	4.98	-
	-	-		-	-	-	-	4.99	-
	-	-	2295.90	-	-	-	-	5.51	-
	-	-		-	-	-	-	5.53	-
	-	-		-	-	-	-	5.54	-
	-	-		-	-	-	-	5.56	-
	-	-		-	-	-	-	5.58	-
Holding									
	-	-	2551.00	-	-	-	-	6.04	-
	-	-		-	-	-	-	6.29	-
	-	-		-	-	-	-	6.30	-
	-	-		-	-	-	-	6.32	-
	-	-		-	-	-	-	6.34	-
	-	-		-	-	-	-	6.35	-
	-	-		-	-	-	-	6.38	-
	-	-		-	-	-	-	6.48	-
	-	-		-	-	-	-	6.58	-
	-	-		-	-	-	-	6.71	-
	-	-		-	-	-	-	6.78	-
	-	-		-	-	-	-	6.83	-
	-	-		-	-	-	-	6.91	-
	-	-		-	-	-	-	6.98	-
	-	-		-	-	-	-	7.01	-

DATE	TIME (Hrs)	PRESSURE GAUGE READING kg/cm²	LOAD IN MT	Dial Gauge				AVERAGE SETTLEMENT IN MM	REMARK
				Reading 1	Reading 2	Reading 3	Reading 4		
	-	-		-	-	-	-	7.03	-
	-	-		-	-	-	-	7.04	-
	-	-		-	-	-	-	7.04	-
	-	-		-	-	-	-	7.05	-
	-	-		-	-	-	-	7.04	-
	-	-		-	-	-	-	7.04	-
	-	-		-	-	-	-	7.03	-
	-	-		-	-	-	-	7.00	-
	-	-		-	-	-	-	6.98	-
	-	-		-	-	-	-	6.96	-
Unloading									
	-	-	2295.90	-	-	-	-	6.93	-
	-	-		-	-	-	-	6.93	-
	-	-		-	-	-	-	6.92	-
	-	-	2040.80	-	-	-	-	6.90	-
	-	-		-	-	-	-	6.88	-
	-	-		-	-	-	-	6.87	-
	-	-	1836.72	-	-	-	-	6.83	-
	-	-		-	-	-	-	6.83	-
	-	-		-	-	-	-	6.83	-
	-	-	1632.64	-	-	-	-	6.77	-
	-	-		-	-	-	-	6.75	-
	-	-		-	-	-	-	6.75	-
	-	-	1428.56	-	-	-	-	6.65	-
	-	-		-	-	-	-	6.64	-
	-	-		-	-	-	-	6.63	-

DATE	TIME (Hrs)	PRESSURE GAUGE READING kg/cm²	LOAD IN MT	Dial Gauge				AVERAGE SETTLEMENT IN MM	REMARK
				Reading 1	Reading 2	Reading 3	Reading 4		
	-	-	1224.48	-	-	-	-	6.51	-
	-	-		-	-	-	-	6.51	-
	-	-		-	-	-	-	6.50	-
	-	-	1020.40	-	-	-	-	6.32	-
	-	-		-	-	-	-	6.31	-
	-	-		-	-	-	-	6.30	-
	-	-	816.32	-	-	-	-	5.98	-
	-	-		-	-	-	-	5.97	-
	-	-		-	-	-	-	5.97	-
	-	-	612.24	-	-	-	-	5.45	-
	-	-		-	-	-	-	5.43	-
	-	-		-	-	-	-	5.43	-
	-	-	408.16	-	-	-	-	5.21	-
	-	-		-	-	-	-	5.21	-
	-	-		-	-	-	-	5.20	-
	-	-	204.08	-	-	-	-	4.87	-
	-	-		-	-	-	-	4.85	-
	-	-		-	-	-	-	4.84	-
	-	-	0.00	-	-	-	-	3.46	-
	-	-		-	-	-	-	3.46	-
	-	-		-	-	-	-	3.45	-

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